

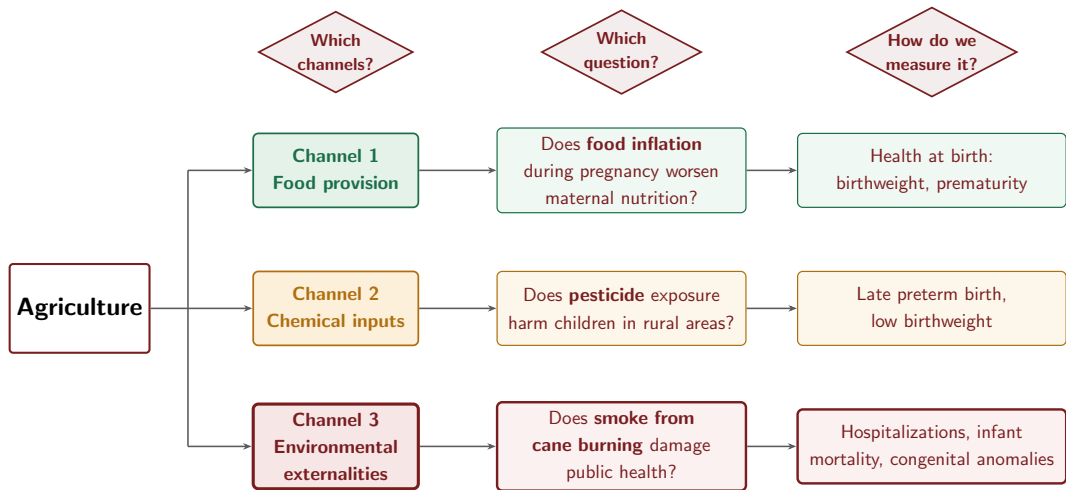
Technical change and health: Evidence from the reduction of sugarcane burning in Brazil

Leonardo Cardoso Liqing Li Deyvid Leite Daiana Wischral David Just

leonardocbcardoso.com



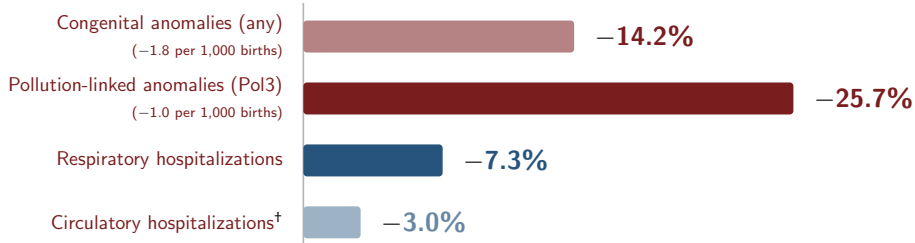
How does agriculture affect health?



Can a technical change in agriculture improve health?

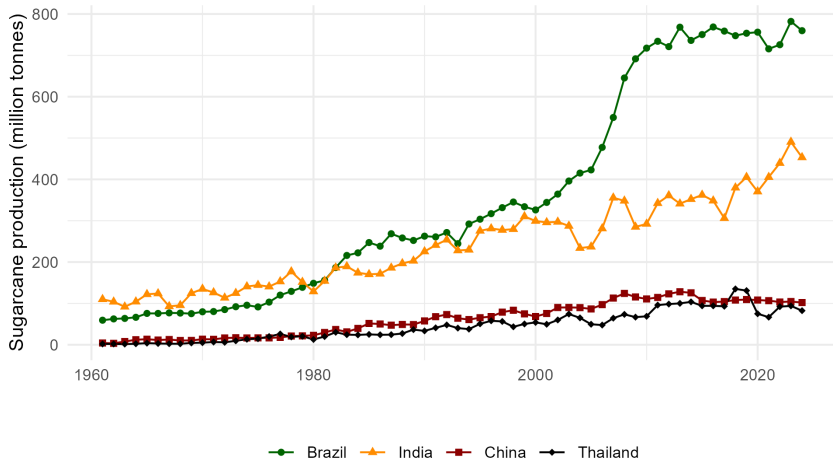


Reduction per one-standard-deviation increase in exposure



And it really is the smoke: ✓ pollution fell **only** in harvest months ✓ **downwind** areas gained more

Brazil Leads World Sugarcane Production



Overclaiming vs. Underselling

São Paulo's shift away from cane burning is, in our view, one of the best settings in the world to study agricultural fire.

- Fast and near-complete: Mechanized harvest
33% (2007) → >95% (2015) [Conab, 2024]
- Intensive: 20% of the state's land under cane by
2010 [IBGE, 2024]
- Not a small player: São Paulo produces 21% of
world sugarcane [FAO, 2025] in a area three
times smaller than Texas



São Paulo (green) over Texas, same scale

Our Source of Identification

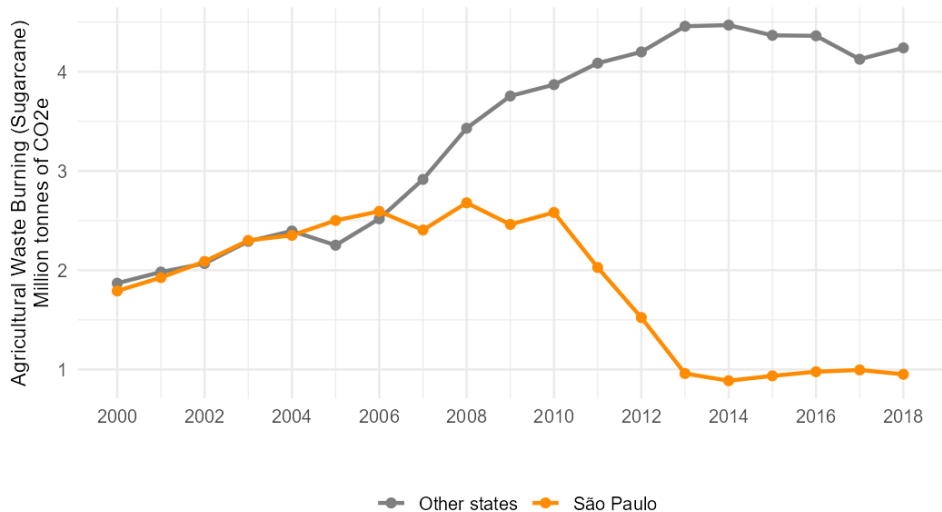
There was a reduction in burning driven by...

- Law
 - ▶ State Law 11.241/2002: gradual ban on straw burning (targets: 50% by 2011, 100% by 2021)
- Protocol
 - ▶ Green Ethanol Protocol (2007): voluntary industry agreement that accelerated deadlines to 2014–2017 in exchange for preferential financing and export market access
- Labor costs
 - ▶ Rising rural wages and export sustainability standards reinforced the shift [Davis, 2017]

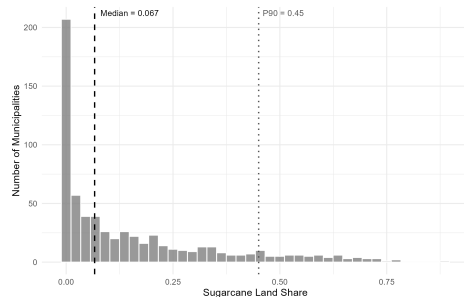
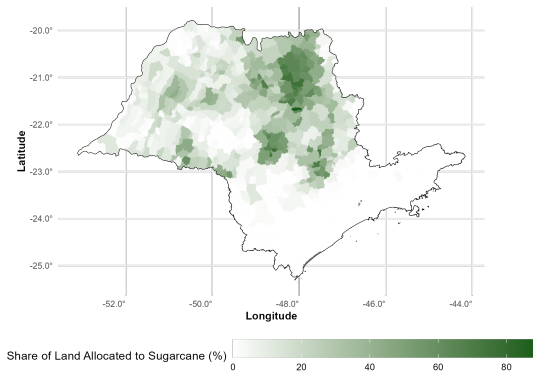
Area allowed to pre-harvest burning



Emissions from sugarcane agricultural waste burning



With spatial heterogeneity in this reduction...



Right-skewed exposure: median 0.067, top decile above 0.45

Combining the timing and the spatial heterogeneity

Difference-in-differences, continuous exposure [Callaway et al., forthcoming]

$$\ln(Y_{it}) = \alpha \cdot \underbrace{Land_i \times D_{t>2010}}_{\text{Tech. Change } (TC_{it})} + \theta_i + \gamma_t + \varepsilon_{it}$$

$Land_i$ = cane land share, 2000–2010 average; $D_{t>2010} = \mathbf{1}[t > 2010]$

Every municipality is treated in proportion to its cane share

- θ_i, γ_t : municipality and year FE; population weights; SEs clustered by municipality
- $SD(Land_i) = 0.19$ — the unit for all “one-SD” effects below

The Literature

- Rangel and Vogl [2019] – Review of Economics and Statistics: wind-based smoke exposure → lower birthweight, shorter gestation (São Paulo)
- Chagas et al. [2016] – Regional Science and Urban Economics: spatial DiD, sugarcane → respiratory hospitalizations, treatment is a binary median-split

Our contribution

- Broader outcomes: circulatory disease, infant mortality, a biologically-grounded congenital anomaly index
- Holding sugarcane production constant, what is the impact of reducing burning?
- Continuous rather than binary exposure, plus three sharper designs (season, wind, dose)

Data Sources

- **Hospitalizations:** SIH/SUS, 2008–2021 (ICD J00–J99, I00–I99), by patient's municipality of residence
- **Births:** SINASC, 2002–2023, $N \approx 12\text{M}$ individual records
- **Mortality:** SIM, 2001–2023, by municipality of residence
- **Treatment:** PAM-IBGE sugarcane land share, 2000–2010 average
- **PM10/PM25:** CAMS global reanalysis (EAC4), 2003–2024
- **Fire:** MODIS MCD64A1 $\times$ MapBiomas LULC (cane pixels), Google Earth Engine
- **Wind:** NASA POWER/MERRA-2 reanalysis, Jul–Sep 2002–2022

All sources are administrative and cover the universe of events — no survey sampling error.

Results

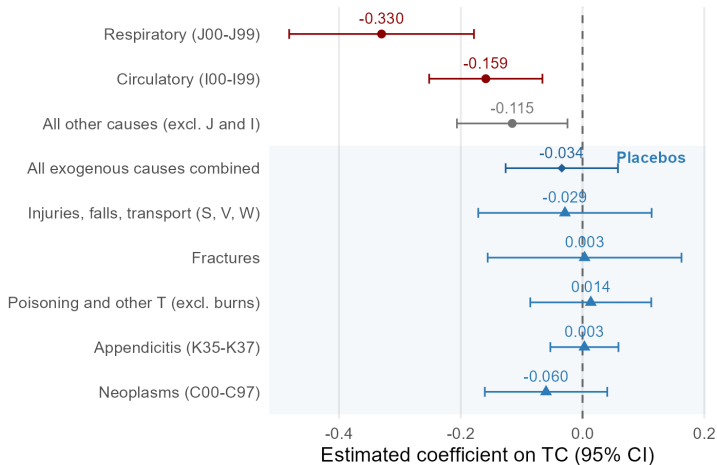
Respiratory Hospitalizations Fall

$y = \ln(\text{hosp. per capita})$	Full (2008–2021)		Pre-Covid (2008–2019)	
	(1)	(2)	(3)	(4)
<i>Panel A: Respiratory (J00–J99)</i>				
Tech. Change	−0.386*** (0.093)	−0.384*** (0.093)	−0.300*** (0.092)	−0.302*** (0.092)
<i>Panel B: Circulatory (I00–I99)</i>				
Tech. Change	−0.159*** (0.047)	−0.161*** (0.047)	−0.156*** (0.043)	−0.157*** (0.043)
log(GDP/cap) control		X		X
Municipality & Year FE	X	X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Pop.-weighted; clustered SEs; by residence.

- **1 SD more exposure** $\Rightarrow$ respiratory $\downarrow$ 7.3%; stable when controlling for GDP/capita
- Circulatory $\downarrow$ 3.0%, but circulatory outcomes were already converging before 2011 — not distinguishable from that pre-existing convergence

Placebo: Outcomes Smoke Cannot Affect



Every placebo is zero; respiratory falls about ten times more (difference -0.30^{***}). Other causes fall a third as much — consistent with broader health gains from cleaner air.

Backup: Dose-Response by Exposure Quartile

Post-2010 $\times$ quartile of cane share	Q1	Q2	Q3	Q4
Mean cane share in quartile	0.005	0.055	0.175	0.465
Respiratory hosp.	0.002 (0.054)	-0.034 (0.057)	-0.138*** (0.049)	-0.171*** (0.045)
Circulatory hosp.	0.018 (0.026)	-0.019 (0.031)	-0.070*** (0.019)	-0.058** (0.029)

The response rises with exposure — no effect where cane is marginal [Callaway et al., forthcoming]. Terciles and quintiles show the same pattern.

Birth Outcomes: Fewer Congenital Anomalies

	Any Anomaly (SINASC flag)	Pol3 Index (NTD+Orofac.+Card.)	LBW (<2,500g)	VLBW (<1,500g)
Tech. Change	-0.009 22* (0.004 86)	-0.005 45** (0.002 29)	0.003 33** (0.001 68)	-0.000 33 (0.000 73)
Baseline mean (post-2010)	12.31/1000	4.02/1000	9.21%	1.44%

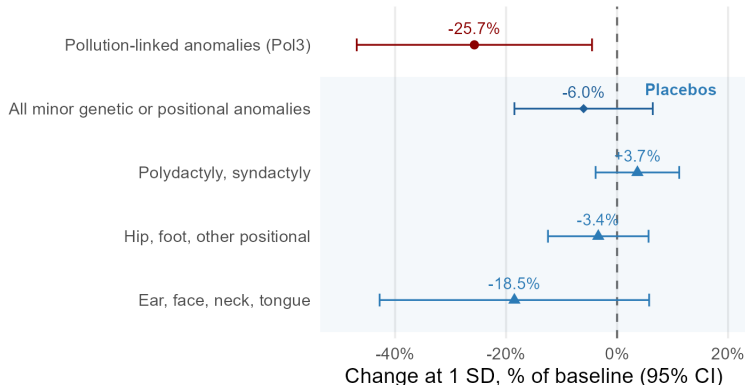
Individual (Sinasc) 2002–2023, $N \approx 12\text{M}$. FE: Mun., year, educ., sex, marital status.

- **Congenital anomalies fall 14.2%**; the **Pol3 pollution-linked index falls 25.7%** at 1 SD of exposure
- **Very low birthweight: null.** Low birthweight carries a small positive coefficient ($p = 0.048$), not corroborated by mean birthweight or VLBW — we do not read it as an effect
- Post-2010 baselines: recorded anomaly rates rise $\approx 10\times$ over 2002–2023 with fetal ultrasound (year FE absorb this trend)

Which Anomalies?

$y = \text{Pr}(\text{anomaly type})$	Neural Tube (Q00–Q07)	Orofacial (Q35–Q37)	Cardiac (Q20–Q26)	Pol3 Index
Tech. Change	−0.000 743*** (0.000 254)	−0.000 122 (0.000 128)	−0.004 724** (0.002 038)	−0.005 445** (0.002 291)
Baseline, post-2010 (per 1,000)	1.15	0.76	2.27	4.02
Relative effect (1 SD)	−12.3%	ns ($p = 0.34$)	−39.5%	−25.7%
Per 1,000 births (1 SD)	−0.14	ns	−0.90	−1.03

Placebo: Anomalies Smoke Cannot Cause



Minor genetic or positional anomalies (extra or fused fingers, clubfoot, hip, ear, tongue) are as common as Pol3 but do not move; the gap is significant ($p < 0.001$). Better prenatal detection in cane areas would have moved both.

Mortality: No Detectable Effect, Respiratory or Circulatory

PPML, 2001–2023	Respiratory		Circulatory	
All ages	0.021	(0.101)	−0.134	(0.081)
Children (<5)	−0.061	(0.138)	−0.126	(0.239)
Elderly (60+)	−0.014	(0.119)	−0.162	(0.089)

Deaths by municipality of residence (SIM), with population (births for children) as exposure. Circulatory coefficients are at most marginal ($p < 0.1$) and circulatory mortality was already converging before 2011 (pre-trend $p < 0.01$).

Mortality is the least sensitive margin: the gains appear in morbidity — hospitalizations and congenital anomalies — not in deaths.

Is It Really the Smoke?

Less Fire and Less Particulate Matter

$$\ln(Y_{it}) = \alpha \cdot Land_i \times D_{t>2010} + \theta_i + \gamma_t + \varepsilon_{it} \quad (\text{same treatment as before})$$

$Y_{it} \in \{ \text{burned area on cane pixels, PM10, PM25} \}$

	Harvest (Jul–Sep)	Off-season (Jan–Mar)
<i>Panel A: Sugarcane burning (MODIS × MapBiomass)</i>		
$\ln(\text{burned area, cane pixels})$	−4.964*** (0.450)	—
<i>Panel B: Particulate matter (CAMS reanalysis)</i>		
$\ln(\text{PM10})$	−0.096*** (0.010)	−0.005 (0.007)
$\ln(\text{PM25})$	−0.094*** (0.010)	−0.001 (0.007)

*** $p < 0.01$. Off-season burned area remains near zero throughout (placebo).

- Cane burned area $\approx -53\%$ at mean exposure; PM10 and PM25 $\downarrow 1.8\%$ at 1 SD — **only during harvest**

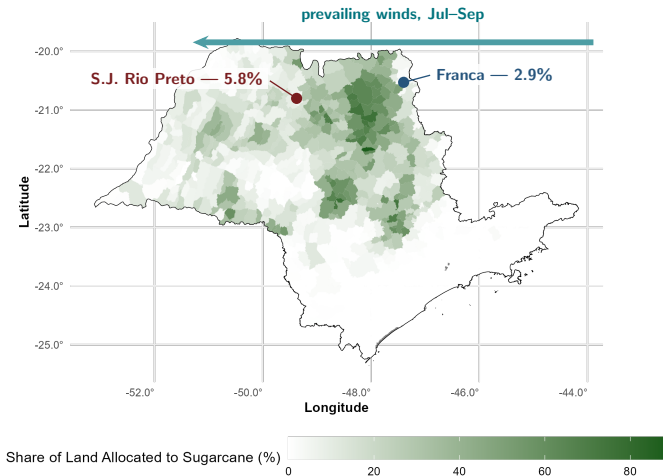
Health Gains Concentrate in Harvest Months

$$\ln Y_{itm} = \alpha \cdot TC_{it} + \beta \cdot (TC_{it} \times Harvest_m) + \theta_{i \times m} + \gamma_t + \varepsilon_{itm}$$

$y = \ln(\text{hosp.}/\text{pop})$	Respiratory	Circulatory
Tech. Change (off-season)	-0.122*** (0.030)	-0.071*** (0.019)
Tech. Change $\times$ Harvest	-0.064*** (0.009)	-0.016 (0.010)

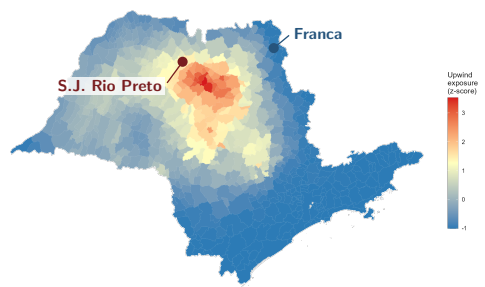
Respiratory hospitalizations fall more when burning would have occurred. Circulatory hospitalizations show no such difference. An income or employment channel from mechanization would not concentrate in harvest months.

Why the wind matters



$$WindExp_{it} = \sum_j \frac{\max(0, \cos(\hat{w}_{it} - \angle_{i \rightarrow j}))}{dist_{ij}} \cdot Land_j$$

Smoke Travels with the Wind

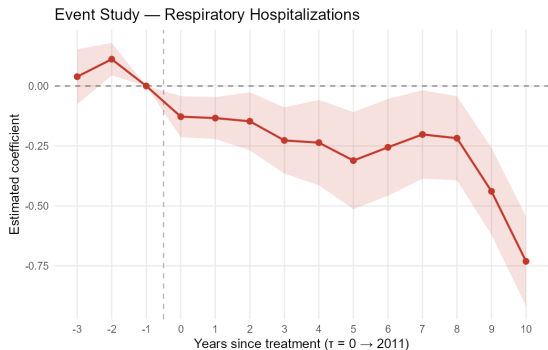


	Respiratory		Circulatory	
	Full	Pre-Cov.	Full	Pre-Cov.
Tech. Change	-0.329***	-0.257***	-0.158***	-0.156***
WindExp	-0.043*	-0.031	-0.031	-0.015

Municipalities downwind of cane fields — e.g. São José do Rio Preto — gain more than upwind ones such as Franca with identical own-exposure. Wind direction is plausibly exogenous to local health trends and, just as importantly, it is what the environmental economics literature now expects.

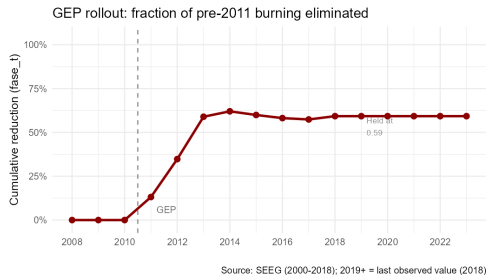
More Robustness

Did Exposed Municipalities Already Trend Differently?



- 2008: near zero. 2009: a positive spike that coincides with the **H1N1 influenza pandemic** — it disappears when 2009 is dropped, and post-treatment estimates do not change
- Post-treatment effects grow as the burning ban phases in
- Circulatory: pre-treatment coefficients decline toward 2010 ($p = 0.04$) — hence our caution on that outcome (backup)

Continuous Treatment Intensity



	Respiratory	Circulatory
Binary (0/1)	−0.330***	−0.159***
Continuous ($\times fase_t$)	−0.576***	−0.218**

Replacing the post-2010 dummy with $fase_t$ — the fraction of burning emissions already eliminated — gives an implied respiratory effect at the mean of -0.323 , essentially the binary estimate (-0.330).

Robustness: Excluding São Paulo City

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Baseline	-0.386***	-0.300***	-0.159***	-0.156***
Excl. SP city	-0.416***	-0.335***	-0.158***	-0.140***

São Paulo city has 12M inhabitants and zero cane share — a heavily weighted, mechanically untreated observation. Dropping it leaves the estimates essentially unchanged.

What Are the Stakes?

- Aggregating the hospitalization reduction over 2011–2021 across exposed municipalities: **≈74,000** fewer respiratory hospital admissions
- At 2019 SUS reimbursement rates: **R\$ 84 million** (USD 21M) in avoided hospitalization costs
- **≈2,504** fewer births with cardiac malformations (2011–2023) $\Rightarrow$ **R\$ 34 million** (USD 9M) in avoided surgical costs

R\$ 118 million (USD 30 million) over a decade — a conservative *lower bound* (it excludes outpatient care, productivity losses, and BPC/LOAS transfers; circulatory hospitalizations, up to R\$ 90 million, are left out). These are health-system savings only: they do not net out the costs of mechanization to growers or to displaced cane cutters.

Takeaway

- A gradual, phased ban on pre-harvest sugarcane burning — backed by regulation *and* voluntary export-market incentives — delivered measurable respiratory and perinatal health gains
- The pattern across outcomes traces a **dose-response gradient**: null mortality at all ages → significant respiratory hospitalizations → significant congenital anomalies — morbidity and fetal development respond before mortality does
- Mechanization also displaced manual cane cutters — a cost we do not measure; the health gains are one side of the ledger
- Relevant beyond São Paulo: Northeast Brazil (Pernambuco: >90% manual), India, Thailand, Florida still rely on pre-harvest burning

Thank you! Questions?

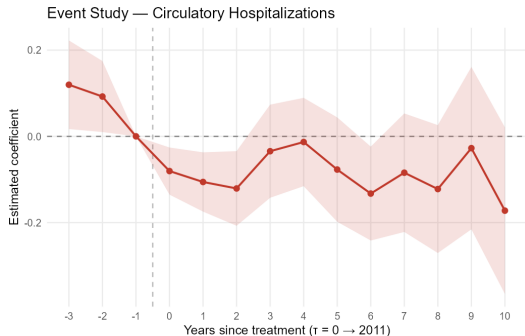
Backup Slides

Summary Statistics

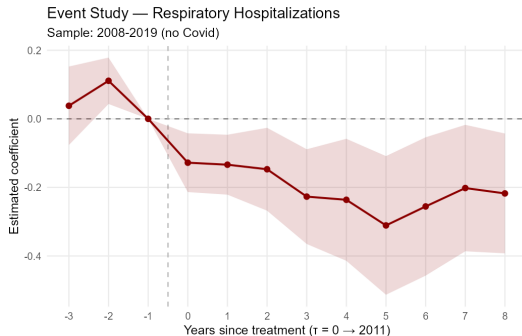
Variable	<i>N</i>	Mean	Median	SD	Max
Sugarcane land share (2000–2010)	642	0.150	0.067	0.190	0.886
Respiratory hosp. per 1,000 pop.	8,988	7.784	—	5.198	46.398
Circulatory hosp. per 1,000 pop.	8,988	8.107	—	3.980	38.560
Any anomaly (per 1,000 births)	11,882	18.662	—	91.807	1,000
Infant resp. mortality (per 1,000 births)	14,765	0.832	—	3.343	111.111

Event Study — Circulatory and Pre-Covid Samples

Circulatory (full sample)

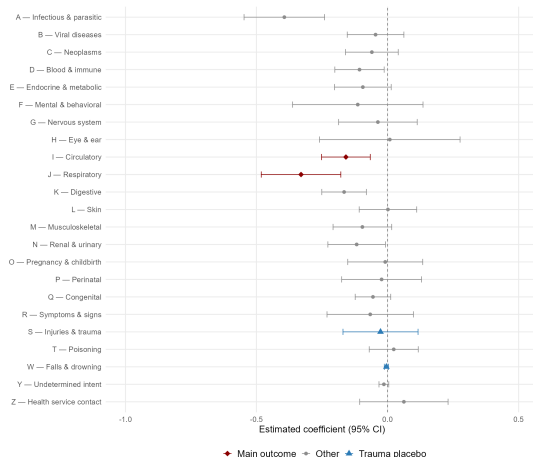


Respiratory (pre-Covid)



Respiratory: the pattern survives dropping the Covid years. Circulatory: pre-treatment coefficients are positive and decline toward 2010 ($p = 0.04$) — the circulatory estimates cannot be distinguished from a pre-existing convergence.

Backup: All ICD-10 Chapters



Placebo chapters — injuries (S) and falls (W) — do not move, and most other chapters are null. Infectious diseases (A) also fall, mostly intestinal infections.

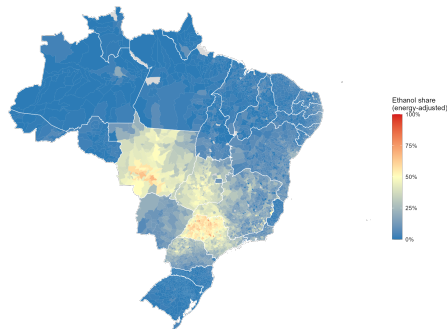
Backup: What About the Labor Channel?

Mechanization displaced manual cane cutters; changes in income or migration could affect health on their own. Four pieces of evidence point to smoke instead:

- **Seasonality:** respiratory effects concentrate in harvest months ($\text{Tech. Change} \times \text{Harvest} = -0.064^{***}$) — an income or employment channel would not be seasonal
- **Wind:** downwind municipalities gain more, holding own exposure fixed (suggestive)
- **Income:** controlling for municipal GDP per capita leaves the estimate unchanged (-0.384 vs. -0.386)
- **Placebos:** hospitalizations for injuries and falls do not move

Composition does shift: births to mothers with 12+ years of schooling fall (-0.034 , $p = 0.03$); the individual birth models control for maternal education.

Backup: Ethanol Market Context



Energy-adjusted hydrated ethanol share in the fuel mix, 2020–2024. Hydrated ethanol competes directly with gasoline where transport costs from the cane belt are low.

Backup: Aggregate Birth Outcomes (Municipality-Year)

	$\ln(\text{birthweight})$	$\ln(\text{LBW})$	$\ln(\text{VLBW})$	$\ln(\text{anomalies})$
Tech. Change	-0.003 (0.004)	0.029 (0.023)	-0.064 (0.058)	-0.844*** (0.319)

The aggregate specification confirms the anomaly result; the individual-level LPM shown earlier additionally decomposes it by diagnostic group.

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